1	**ACSBR/2025/I/02**
	(a) Factorise $x^3 - y^3$. [2] **(b)** Hence express $\dfrac{7^{\frac{3}{2}} - 2^{\frac{3}{2}}}{\sqrt{7} - \sqrt{2}}$ in the form $a + b\sqrt{14}$, where a and b are integers. [3]
2	**ACSBR/2025/I/05**
	Express $\dfrac{2x^3 - 2x^2 - 24x - 7}{x^2 - x - 12}$ in partial fractions. [5]
3	**ACSBR/2025/II/04**
	It is given that $f(x) = x^3 + mx^2 + 6x - 4$, where m is a constant. **(a)** Show that $m = 6$ given that $x + 2$ is a factor of $f(x)$. [2] **(b)** Using the value of m in **part (a),** solve the equation $f(x) = 0$, expressing non-integer roots in surd form. [4]
4	**BASS/2025/I/03**
	(a) Explain why $\dfrac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21}$ is not a proper fraction. [1] **(b)** Express $\dfrac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21}$ in partial fraction. [6]
5	**BASS/2025/II/02**
	The remainder when the expression $2hx^3 + kx^2 + 4x + 4$ is divided by $(x - 2)$ is four times the remainder when it is divided by $(x + 1)$. **(a)** Given that $(x - 1)$ is a factor of the expression, find the value of h and of k. [4] **(b)** Hence solve $2hx^3 + kx^2 + 4x + 4 = 0$, leaving your answers in the form $a \pm b\sqrt{3}$ where applicable. [4]
6	**GMS/2025/II/10**
	The cubic polynomial $f(x)$ is such that the coefficient of x^3 is -1. The roots of the equation $f(x) = 0$ are 1, 2 and k. Given that $f(x)$ has a remainder of 8 when divided by $(x - 3)$, find **(a)** the value of k, [3] **(b)** the remainder when $f(x)$ is divided by $(x + 3)$. [1]

1	(a)	$(x-y)(x^2+xy+y^2)$
	(b)	$\frac{(\sqrt{7})^3-(\sqrt{2})^3}{\sqrt{7}-\sqrt{2}}$ $=\frac{(\sqrt{7}-\sqrt{2})(7+\sqrt{14}+2)}{\sqrt{7}-\sqrt{2}}$ $=9+\sqrt{14}$
2	$\frac{2x^3-2x^2-24x-7}{x^2-x-12}$ $=2x+\left(\frac{-7}{x^2-x-12}\right)$ $=2x+\left(\frac{-7}{(x-4)(x+3)}\right)$ $\frac{-7}{(x-4)(x+3)}=\frac{A}{x-4}+\frac{B}{x+3}$ $-7=A(x+3)+B(x-4)$ $x=4$: $-7=7A$ $A=-1$ $x=-3$; $-7=-7B$ $B=1$ $\frac{2x^3-2x^2-24x-7}{x^2-x-12}=2x-\frac{1}{x-4}+\frac{1}{x+3}$	

3	(a)	$f(-2) = -8 + 4m - 12 - 4$ $4m - 24 = 0$ $m = 6$ (shown)
	(b)	$$\begin{array}{r} x^2 + 4x - 2 \\ (x+2)\overline{)x^3 + 6x^2 + 6x - 4} \\ -x^3 + 2x^2 \\ 4x^2 + 6x \\ 4x^2 + 8x \\ -2x - 4 \\ -2x - 4 \end{array}$$ $f(x) = (x+2)(x^2 + 4x - 2)$ $x + 2 = 0$ $x = -2$ $x^2 + 4x - 2 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-4 \pm \sqrt{4^2 - 4(1)(-2)}}{2(1)}$ $= \frac{-4 \pm \sqrt{24}}{2}$ $x = -2 \pm \sqrt{6}$

4	a	Degree of numerator = 3, Degree of denominator = 2 Since degree of numerator $>$ degree of denominator, $\dfrac{x^3+5x^2-14x-10}{x^2+4x-21}$ is not proper fraction
	b	$\begin{array}{r\|l} & x \quad + \quad 1 \\ \hline x^2+4x-21 & x^3+5x^2-14x-10 \\ & -(x^3+4x^2-21x) \\ \hline & \qquad x^2+7x-10 \\ & \quad -(x^2+4x-21) \\ \hline & \qquad\quad 3x+11 \end{array}$ $\dfrac{x^3+5x^2-14x-10}{x^2+4x-21}=x+1+\dfrac{3x+11}{x^2+4x-21}$ Let $\dfrac{3x+11}{(x+7)(x-3)}=\dfrac{A}{x+7}+\dfrac{B}{x-3}$ $3x+11=A(x-3)+B(x+7)$ When $x=-7$, $3(-7)+11=A(-7-3)+B(-7+7)$ $-10=-10A$ $A=1$ When $x=3$, $3(3)+11=A(3-3)+B(3+7)$ $20=10B$ $B=2$ Hence, $\dfrac{x^3+5x^2-14x-10}{x^2+4x-21}=x+1+\dfrac{1}{x+7}+\dfrac{2}{x-3}$.
5	**a**	$2h(2)^3+k(2)^2+4(2)+4=4[2h(-1)^3+k(-1)^2+4(-1)+4]$ $16h+4k+12=-8h+4k$ $h=-\dfrac{1}{2}$ Since is a factor, $2\left(-\dfrac{1}{2}\right)(1)^3+k(1)^2+4(1)+4=0$ $k=-7$

	b	$2hx^3 + kx^2 + 4x + 4 = 0$ $-x^3 - 7x^2 + 4x + 4 = 0$ $x^3 + 7x^2 - 4x - 4 = 0$ $x = 1 \left\lvert \begin{array}{cccc} 1 & 7 & -4 & -4 \\ \downarrow & 1 & 8 & 4 \\ \hline 1 & 8 & 4 & 0 \end{array} \right.$ $(x-1)(x^2+8x+4) = 0$ $(x-1) = 0 \quad \text{or} \quad x^2+8x+4 = 0$ $x^2+8x+4=0$ $x = \dfrac{-8 \pm \sqrt{64-16}}{2}$ $= \dfrac{-8 \pm 4\sqrt{3}}{2}$ $= -4 \pm 2\sqrt{3}$ $\therefore \quad x = 1, \quad -4+2\sqrt{3} \quad \text{or} \quad -4-2\sqrt{3}$
6	**(a)** $f(x) = 0$ $(x-1)(x-2)(x-k) = 0$ Since coefficient of $x^3 = -1$ $f(x) = -(x-1)(x-2)(x-k)$ $f(3) = 8$ $-(3-1)(3-2)(3-k) = 8$ $3-k = -4$ $k = 7$ **(b)** $f(-3) = -(-3-1)(-3-2)(-3-7)$ $= 200$	